



S2.3 The metallic model

Structure 2 · Models of bonding and structure · IB Diploma · Concept guide and original exam-style practice

Understand the concept

Metallic bonding consists of a lattice of closely packed metal cations surrounded by a 'sea' of delocalised valence electrons that are free to move throughout the whole structure, with the electrostatic attraction between the cations and the delocalised electrons holding the structure together. This model explains characteristic metallic properties: electrical conductivity (delocalised electrons carry charge and can move in any direction, in solid or liquid state), malleability and ductility (the non-directional bonding allows layers of cations to slide past each other without breaking bonds), and generally high melting points (which increase with a greater number of delocalised electrons per atom, smaller cation radius, and greater cation charge, all strengthening electrostatic attraction).

Worked example: follow the reasoning

Explain why magnesium has a higher melting point than sodium.

Step 1: Both are metals with delocalised electron/cation lattice bonding.

Step 2: Mg atoms contribute 2 delocalised electrons per atom (from the $3s^2$ electrons) to the 'sea', whereas Na contributes only 1 (from $3s^1$).

Step 3: Mg^{2+} also has a greater ionic charge and a smaller ionic radius than Na^+ .

Answer: Mg has stronger metallic bonding than Na because of a greater density of delocalised electrons and a higher cation charge with smaller radius, giving stronger electrostatic attraction between cations and electrons, so more energy is required to melt Mg, giving it a higher melting point.

Quick check · multiple choice

Which property of metals is best explained by the presence of delocalised electrons free to move through the lattice?

- A High density
- B Brittleness
- C Electrical conductivity
- D Low reactivity

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

- 1 (a) Describe the structure and bonding in a typical metal such as aluminium. [2] [7]
(b) Explain, using the metallic bonding model, why metals are malleable while ionic solids are brittle. [3]
(c) An alloy of copper and tin (bronze) is harder than pure copper. Suggest, in terms of structure, why alloying increases hardness. [2]

Final answer: _____



Answers and explanation

Question 1 [7]

- (a) A giant lattice of closely packed metal cations (Al^{3+}) is surrounded by a 'sea'/delocalised cloud of valence electrons that are free to move throughout the whole structure; electrostatic attraction between the cations and the delocalised electrons holds the structure together. [2]
- (b) In metals, the delocalised electron 'glue' is non-directional, so layers of cations can slide over each other into new positions while remaining surrounded by delocalised electrons, and the metallic bonding is not broken — this allows malleability. In ionic solids, a shearing force shifts layers so that ions of the same charge become adjacent, causing strong repulsion, which shatters the lattice (brittle) rather than allowing layers to slide. [3]
- (c) The tin atoms are a different size from copper atoms, so they distort the regular layered arrangement of the copper lattice; this disrupts the regular planes of atoms, making it harder for layers to slide over one another, so the alloy is harder than pure copper. [2]

Quick check answer

C. Delocalised electrons are mobile charge carriers that can move throughout the metallic lattice under an applied potential difference, giving metals high electrical conductivity.

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or content error, then retry the practice question without looking at the solution.

Based on the official IB Chemistry HL/SL subject guide.

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